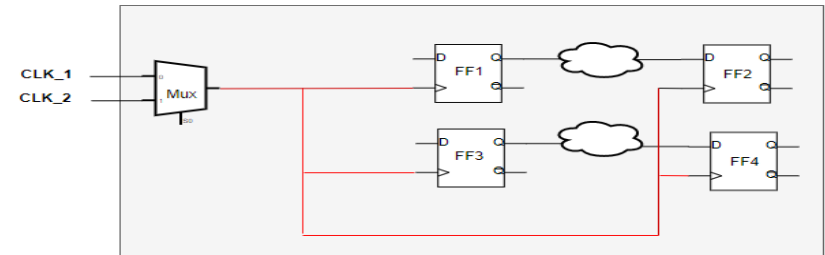


Design Compiler Timing Exception Constraints

3. **Exclusive Clocks** : Two clocks are exclusive if they don't interact with each other. Circuit might multiplex two different clock signals onto a clock line. For examples: **One fast clock for normal operation and the other a slow clock for low power operation.** Or **one clock for functional mode and the other scan clock for ATPG mode.** To prevent the tool from spending time analyzing the interaction between exclusive clocks.
- SDC Command : `set_clock_groups -logically_exclusive -group {clock_list}`
 - SDC Command : `set_clock_groups -physically_exclusive -group {clock_list}`

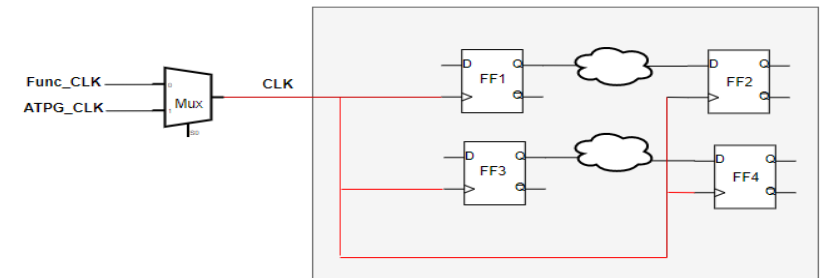
3.1) Logically exclusive Clocks

- The clocks can **physically exist** at the same time, but the design logic guarantees that **only one of them** can reach the functional logic. So STA engineer should not analyze paths between registers clocked by CLK_1 and registers clocked by CLK_2, because those paths cannot be active simultaneously in the functional mode.
- SDC Command :-
`set_clock_groups -logically_exclusive -group {get_clocks CLK1} \`
`-group {get_clocks CLK2}`



3.2) Physically exclusive Clocks

- The clocks **cannot physically exist/toggle** at the same time.
- SDC Command :-
`set_clock_groups -physically_exclusive -group {get_clocks CLK1} \`
`-group {get_clocks CLK2}`



Design Compiler Timing Exception Constraints

3) Max/Min Delay Paths:-

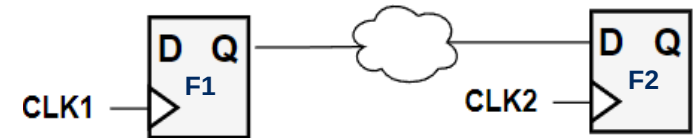
- In **Synopsys Design Compiler (DC)** and **Static Timing Analysis (STA)**. Normally, timing requirements are derived from the relationship between clocks. However **set_max_delay** and **set_min_delay** are path-based timing constraints used to explicitly control the allowable delay of selected timing paths. It is typically used for **Special interface timing requirements**, **Input to Output timing paths** with **non-standard timing requirements** and **Controlling delay across selected Asynchronous CDC paths**
- **SDC Command:**

```
set_max_delay <delay> -from <start_point> -to <end_point>
```

```
set_min_delay <delay> -from <start_point> -to <end_point>
```

3.1) set_max_delay

- It is used to define the **maximum allowed delay** for selected timing paths
- **Examples:-**



- 1) Constrain the path starting from FF1 and ending to FF2 with **minimum** allowable delay with 1 ns

```
set_max_delay 2 -from [get_cells FF1*] -to [get_cells FF2*]
```

- 2) Constrain all paths starting from a flop with launch clock “CLK1” and ending with a flop with capture clock “CLK2” with maximum allowable delay with 2 ns

```
set_max_delay 2 -from [get_clocks CLK1] -to [get_clocks CLK2]
```

Design Compiler Timing Exception Constraints

3.1) **set_min_delay**

- It is used to define the **minimum allowed delay** for selected timing paths
- **Examples:-**

1) Constrain the path starting from FF1 and ending to FF2 with minimum allowable delay with 1 ns

```
set_max_delay 1 -from [get_cells FF1*] -to [get_cells FF2*]
```

2) Constrain all paths starting from a flop with launch clock “CLK1” and ending with a flop with capture clock “CLK2” with maximum allowable delay with 1 ns

```
set_max_delay 1 -from [get_cocks CLK1] -to [get_clocks CLK2]
```